MOOST: A Majority-Oriented Over-Sampling Technique for Aero-Engine Anomaly Detection

Abstract

To address the challenges of enhancing feature learning capabilities from highly imbalanced industrial data and achieving superior generalization performance, this paper develops a novel deep neural network architecture, namely the deep self-attention feature augmentation network (DSAFAN). The majority-oriented oversampling technique (MOOST) is inserted into the deep learning architectures as an auxiliary tool for minority-class feature augmentation. Considering that it is extremely difficult to enhance the completeness of the minority-class data, the developed MOOST generates out-of-distribution synthetic data with abnormal characteristics under the guidance of normal data, thereby effectively expanding the advantageous region occupied by existing anomalies. Furthermore, the DSAFAN leverages multi-head self-attention mechanism as the feature extraction module, significantly improving the temporal feature learning capability for complex data. Finally, the effectiveness of the proposed methods has been validated through comprehensive experiments involving the real-world engineering data.

Keywords:

Aero-engine anomaly detection; inter-class imbalance; inter-class overlap; majority-oriented oversampling technique; deep self-attention feature-augmentation network

1 Introduction

As the key power plant of aircraft, the health status of aero-engines directly affects the aircraft's flight safety[1]. Early anomaly identification can help adjust the operation and maintenance strategy for aero-engines in a timely manner and reduce the risk of them evolving into catastrophic situations, thereby mitigating flight delays and reducing operational costs[2]. Consequently, there is an urgent need to develop and implement an effective aero-engine anomaly detection framework to ensure the reliable operation of airlines[3].

In recent years, data-driven anomaly detection methods have made significant progress in industry applications[4],[5]. They can be mainly divided into two categories: machine learning-driven anomaly detection methods (MLDADM) and deep learning-driven anomaly detection methods (DLDADM). Traditional MLDADMs tend to exhibit poor feature learning capabilities and face difficulties in directly processing raw data, often relying on manually selected statistical features. If the selected statistical features do not accurately represent the health status of aircraft engines, the MLDADMs may face the risk of misclassification[6]. On the contrary, DLDADMs possess powerful feature learning capabilities and can directly learn discriminative features from raw sensor data, effectively avoiding the aforementioned risks[7],[8],[9].

It is worth noting that DLDADMs typically contain a large number of trainable weight parameters and require sufficient, class-balanced training samples[10]. However, in practical engineering applications, the collection of abnormal samples is often extremely limited compared to that of normal samples, primarily due to the fact that aero-engine faults typically occur under exceptionally rare conditions. In this case, DLDADMs tend to become biased towards the majority-class samples (normal samples) and do not learn sufficient information from the minority-class samples (abnormal samples)[11]. The reason is that classification accuracy of DLDADMs is dominated by normal samples; even if abnormal samples are misclassified as normal ones, the overall classification accuracy can remain relatively high. Consequently, class-imbalanced datasets pose a significant challenge to DLDADMs[12].

Thus far, several studies have been conducted in-depth research on the classification problem of imbalanced datasets, among which over-sampling techniques remain the most widely used solution[13]. Over-sampling techniques increase the number of abnormal samples by copying/synthesizing them to match the number of normal samples. Compared to down-sampling techniques that remove normal samples, over-sampling can help avoid the loss of normal information. As one of the most representative of the several existing over-sampling techniques, the synthetic minority oversampling technique (SMOTE) involves randomly interpolating on the line connecting adjacent minority-class samples to synthesize new ones[14]. However, direct augmentation of the aero-engine dataset using SMOTE-based over-sampling techniques can

encounter the following two challenges:

Highly inter-class overlap: Early-stage abnormal samples exhibit minimal deviation from normal samples; that is, they substantially overlap with the normal data. Direct data augmentation in the original data space may result in the synthesized data falling into the danger zone, where the number of normal samples is more than that of abnormal samples. In such case, synthesized samples will be closer in characteristics to normal data, but will carry labels of abnormal data (hereafter, such data will be referred to as noise data). These data may severely mislead the training of the DLDADMs[15].

Incomplete abnormal data: In engineering practice, the abnormal data that can be gathered from aero-engines are extremely scarce and incomplete, comprising only a small fraction of all possible abnormal modes. Traditional data augmentation strategies can only synthesize in-distribution data, and are inherently unable to generate out-of-distribution anomalies near decision boundaries, thereby limiting their ability to improve the generalization performance of the DLDADMs[16],[17].

Aiming at these challenges, we develop a novel supervised anomaly detection method for aero-engines based on the deep self-attention feature augmentation network (DSAFAN). Firstly, the DSAFAN maps the raw data with inter-class overlap to an inter-class separable feature space, thus relieving the problem of inter-class overlap. Subsequently, in the feature space, the majority-oriented over-sampling technique (MOOST) is designed to synthesize abnormal data outward under the guidance of the relatively complete normal data, which can balance the loss generated by inter-class samples. The MOOST can not only reduce the probability of the synthesized data falling into the danger zone but also extend the advantageous areas occupied by abnormal data, making it suitable for anomaly detection under incomplete conditions. What's more, to enhance the feature learning ability of the DSAFAN, the multi-head self-attention mechanism is introduced to capture the complex temporal features from monitoring data.

To sum up, the major contributions are summarized as follows.

- A novel supervised anomaly detection framework is proposed, which not only suppresses the inter-class overlap but also balances the loss difference between minority-class samples and majority-class samples, thereby addressing the challenge of inter-class imbalance.
- The MOOST can be seamlessly integrated into deep learning architectures as the feature augmentation module, in order to effectively eliminate the biased learning in imbalanced scenarios.
- Under the guidance of the relatively complete majority-class data, the MOOST synthesizes minority-class data outward to extend their advantageous zones, thereby enhancing the generalization capability of deep architectures.

• Comprehensive experimental evaluations on real-world aero-engine operation data demonstrate the superior performance of the proposed DSAFAN.

The remainder of this paper is organized as below. Section 2 provides a brief overview of the related work, and Section 3 gives a detailed elaboration on the developed DSAFAN. Experimental comparisons are given in Section 4, and Section 5 concludes this paper.

2 Related Works

2.1 Anomaly detection for aircraft engines

At present, anomaly detection methods for aero-engines are mainly categorized into two types: model-driven anomaly detection methods and data-driven anomaly detection methods. Model-driven anomaly detection methods have good interpretability, but require a significant amount of domain knowledge. Especially for highly complex thermal machinery such as aero-engines, it is extremely difficult to construct an accurate physical analytical model to analyze the fault propagation process and dynamic damage response, which limits their practical application[18]. For example, Simon et al. determine aero-engine faults by calculating the residual between the measured values of the aero-engine and the values predicted by the baseline performance model[19]. In contrast, data-driven anomaly detection methods can automatically learn useful and informative features from operational data and be used to infer the aero-engine health status[20]. Airlines have collected large amounts of engine status data, providing significantly more favorable conditions for the implementation of data-driven methods. Moreover, in recent years, various data-driven methods have emerged and achieved encouraging diagnostic results in anomaly detection[21], including Gaussian mixture models[22], isolated forests[23], and artificial neural networks[24]. For example, Kang et al. construct an anomaly detection framework based on distillation learning for aero-engines and demonstrate efficient detection capabilities of the Vision Transformer. Shang et al. use the superpixel perception graph neural network to extract features from aero-engine blade data and achieve superior performance compared with the state-of-the-art methods. Niu et al. propose a simple class-agnostic feature adaptation method for anomaly detection of aero-engine blade and achieve accurate detection performance even under different class numbers of normal data. However, as highly reliable and safe thermal machinery, aero-engines generally need to undergo several flight cycles to transition from a normal to an abnormal state, resulting in limited data for the latter. In such cases, traditional data-driven methods tend to be biased toward normal samples during training, resulting in poor diagnostic performance for abnormal samples.

2.2 Traditional re-sampling methods for imbalanced classification

The DLDADMs often focus more on minimizing the classification error of majority-class samples when dealing with imbalanced data, while ignoring that of the minority-class samples. In this case, the minority-class samples can be easily misclassified as belonging to the majority class[28]. Data re-sampling is a powerful tool used to address the aforementioned class imbalance issues. Existing data re-sampling methods can be mainly divided into two categories: down-sampling methods and over-sampling methods[29]. Furthermore, down-sampling methods can be divided into random down-sampling methods and heuristic down-sampling methods. Random down-sampling methods randomly remove a certain number of majority-class original samples from the training set. However, the removed samples can contain potentially useful information, which can negatively impact the diagnostic performance of the classification model[30]. To maximally preserve useful information, several heuristic down-sampling methods have been proposed. These methods typically consider the data distribution and remove the majority-class original samples according to certain criteria[31]. For example, Tomek develops two new variants of the shrinkage nearest neighbor rule (Tomek link) to remove majority-class samples in the boundary region based on the distance between sample pairs[32]. Moreover, operation data can be often classified into different clusters using unsupervised clustering algorithms, and the samples closer to the cluster center are typically closer to all samples within the cluster, and hence, contain more representative information. Following this, Lin et al. propose a concise and efficient down-sampling algorithm based on unsupervised clustering, which removes majority-class samples near the cluster edge based on a radial distance threshold, thereby reducing the risk of removing representative data[33].

Similarly, over-sampling methods can also be divided into two categories: random over-sampling methods and heuristic over-sampling methods. Random over-sampling methods involve copying a certain number of minority-class original samples and adding them to the training set. However, this lack of diversity in the randomly replicated samples can easily lead to overly specific classification boundaries during the learning of classification models and lead to over-fitting. To alleviate this problem, various heuristic over-sampling methods have been proposed that guide the synthesis of minority-class samples according to certain criteria. For example, Chawla proposes SMOTE to generate new minority samples by randomly sampling the connection between each minority-class sample and its adjacent samples[14]. It is noteworthy that the SMOTE did not consider the degree of overlap between the minority-class samples and their adjacent majority-class samples. Randomly synthesizing minority-class samples without appropriately considering the majority-class distribution may result in the risk of generating noisy data. To this end, Han et al. propose borderline-SMOTE to synthesize minority-class samples near

the decision hyperplane based on the proportion of adjacent majority-class samples[34]. Bunkhumpornpat et al. develop the safe-level SMOTE, which synthesizes minority samples in a safe area by staying far away from the decision hyperplane[35]. Douzas et al. employ k-means clustering to first divide the input data into different clusters and synthesize the minority-class samples using SMOTE separately for each cluster[36].

3 Proposed anomaly detection framework

This section first discusses the research motivation of this study, followed by the process of constructing the novel aero-engine anomaly detection framework based on the DSAFAN. Finally, the central idea of the developed MOOST is described in detail.

3.1 Research motivations

Anomaly detection of aero-engines is an extremely important process to ensure customer safety and save operating costs for airlines. However, the existing anomaly detection methods face the following challenges.

Inter-class imbalance: Traditional anomaly detection methods typically use cross-entropy loss functions as the optimization objectives. Compared with the extremely scarce and incomplete abnormal data, these methods tend to focus more on the majority-class (normal) data during training. Specifically, even if the cross-entropy loss incurred by an individual abnormal sample may be relatively large, the total cross-entropy loss contributed by all abnormal samples remains negligibly small due to their scarcity. During the optimization process, abnormal information may be overlooked. On the contrary, even if the cross-entropy loss generated by a single normal sample is relatively small, the total cross-entropy loss contributed by all normal samples dominates due to the large sample size. These lead to DLDADMs being more inclined to accurately identify normal operating samples, with their ability to detect abnormal samples remaining poor.

Inter-class overlap: Especially in the early stages of anomalies, the changes in operation data may be too weak to clearly distinguish between normal and abnormal states, i.e., the normal and abnormal samples in the raw space will highly overlap and be difficult to differentiate. In this case, direct data augmentation may generate noisy data that can mislead the training of the DLDADMs and limit the final gain effects.

Complex temporal patterns: The aero-engine operation data is a type of multidimensional time series data, and the relationships between the performance parameters are extremely complex. It is significantly difficult for traditional DLDADMs to directly extract meaningful abnormal patterns from multidimensional time-series datasets.

3.2 General procedure of the proposed anomaly detection framework

To address the imbalance issue under inter-class overlap and minority-class incompleteness conditions, we propose a novel anomaly detection framework based on the DSAFAN. As depicted in Fig. 1, the overall procedure consists of three main modules.

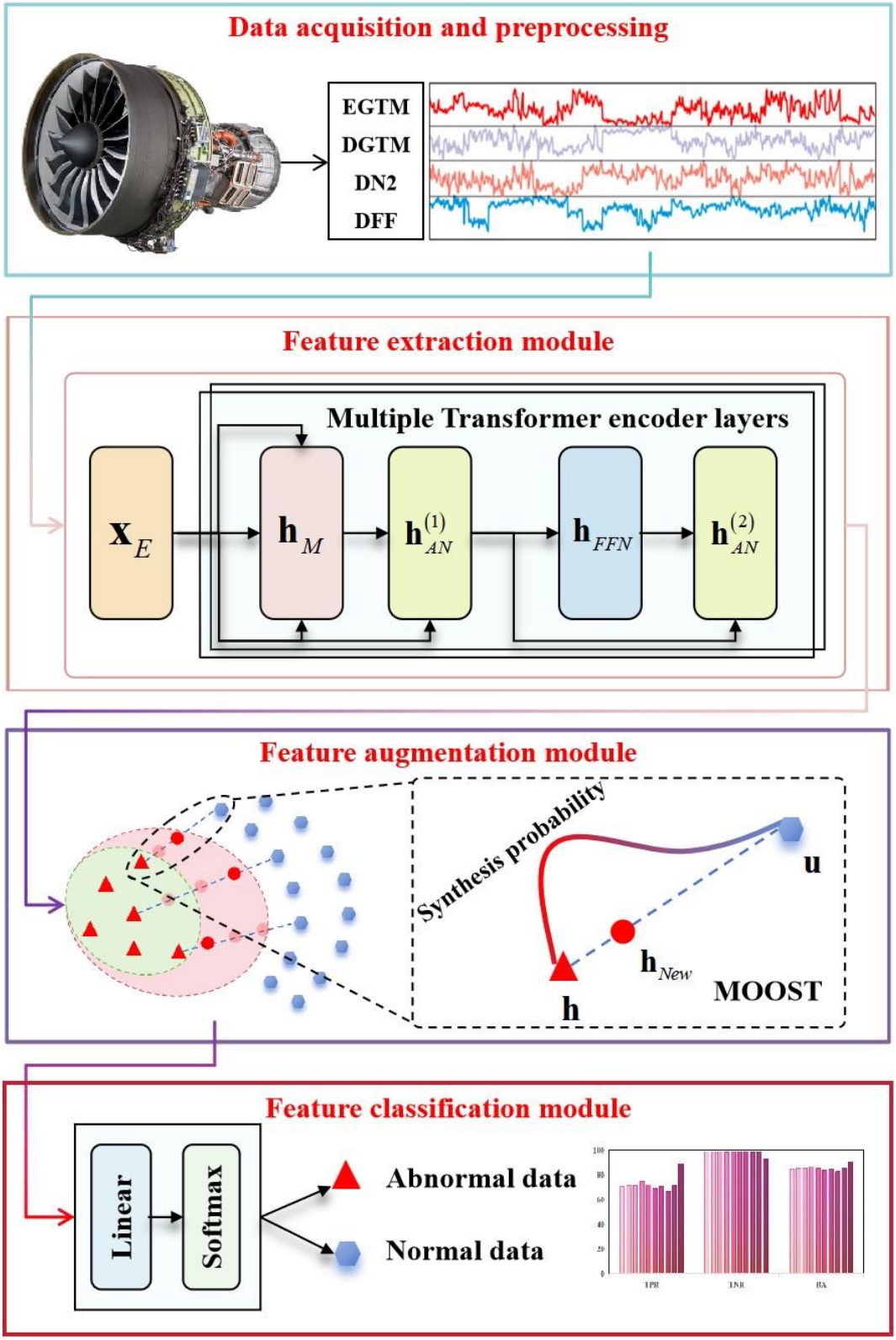


Fig.1 Working principle of the aero-engine supervised anomaly detection method based on the DSAFAN

Feature extraction module: This module is applied to extract more distinguishable anomalous features by incorporating the multi-head self-attention mechanism, while effectively mitigating overlap between normal and abnormal data through mapping raw data into an inter-class separable feature space.

Feature augmentation module: This module is designed to generate synthetic abnormal features, in order to enhance the completeness of the abnormal data and reduce the risk of introducing noise. Specifically, this module synthesizes abnormal features within the inter-class separable feature space, which are subsequently integrated with the original features to assemble a balanced training set.

Feature classification module: This module handles abnormal pattern recognition via a Softmax classifier (SMC). It is trained on the balanced dataset generated by the feature augmentation module, thus learning an unbiased decision boundary to achieve accurate aero-engine anomaly detection.

3.3 Feature extraction for multidimensional time-series data

The goal of the feature extraction module is to capture distinguishable anomalous features while suppressing irrelevant information. Considering the characteristics of aero-engine operation data, this module adopts the Transformer encoder layer as the backbone[37],[38]. As shown in Fig. 2, the Transformer encoder layer consists of 1 multi-head self-attention mechanism, 2 short-cuts & layer normalizations, and 1 fully connected feed-forward layer. Each network unit serves a specific function, as outlined in the following sections.

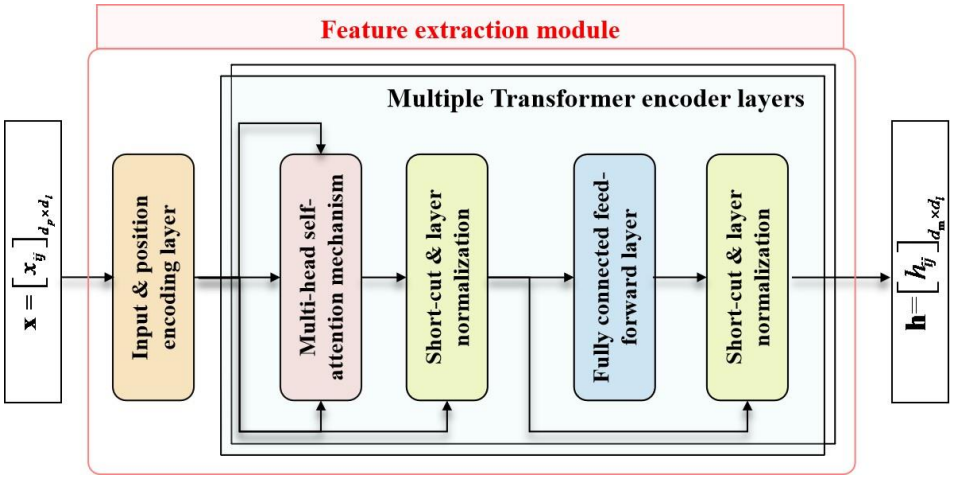


Fig.2 Schematic diagram of the feature extraction module

Input & position encoding layer: The input & position encoding layer can be regarded as the pre-processor of the DSAFAN, and uses one-dimensional convolution to encode the input data. The reason for choosing the convolution operation is that it can efficiently capture specific temporal patterns when processing time series[39], thereby improving the module's ability to extract complex temporal features. The value encoding $\mathbf{x}_{VE}$ of the input data $\mathbf{x}=\left[x_{ij}\right]_{d_p\times d_l}$

(where d_p is the sequence dimension of the input data and d_l is the sequence length of the input data) can be represented as

$$\mathbf{x}_{VE} = \mathrm{Conv1d}\left(\mathbf{x}\right) = \left[x_{ij}^{VE} \right]_{d_{\mathrm{m}} \times d_l}$$

where $\mathrm{Conv1d}\left(\bullet\right)$ represents the one-dimensional convolution operation and d_m is the sequence dimension of the value encoding $\mathbf{x}_{VE}$.

Since the Transformer encoder layer does not consider the position information, it is necessary to add the position information into the value encoding of the input[37]. The position encoding of the input data, $\mathbf{x}_{PE}$, can be expressed as

$$\mathbf{x}_{PE} = \mathrm{PE}\left(\mathbf{x}\right) = \left[x_{ij}^{PE} \right]_{d_{\mathrm{m}} \times d_l}$$

$$x_{ij}^{PE} = \begin{cases} \sin\left(\dfrac{i}{10000^{\frac{j}{d_m}}} \right), & j \bmod 2 = 0 \\ \cos\left(\dfrac{i}{10000^{\frac{j-1}{d_m}}} \right), & j \bmod 2 = 1 \end{cases}$$

where $\mathrm{PE}\left(\bullet\right)$ is the position encoding operation and mod is the withdrawal operation.

Consequently, the final output of the input data through the input & position encoding layer can be expressed as

$$\mathbf{x}_E = \mathrm{Dropout}\left(\mathbf{x}_{VE} + \mathbf{x}_{PE}\right)$$

where $\mathrm{Dropout}\left(\bullet\right)$ is a popular means to prevent over-fitting in deep learning.

Multi-head self-attention mechanism: To enhance the capability of extracting critical information, the multi-head attention mechanism is introduced into the DSAFAN[40], consisting of m_H parallel self-attention layers (shown in Fig.3). Different self-attention layers focus on the important temporal patterns in different mapping sub-spaces and subsequently, capture richer temporal features through fusion. Taking the i-th self-attention layer as an example, $\mathbf{x}_E$ undergoes three linear mappings to obtain the query vector ($\mathbf{Q}_i$), key vector ($\mathbf{K}_i$), and value vector ($\mathbf{V}_i$), which can be expressed as

$$\begin{cases} \mathbf{Q}_i = \mathbf{x}_E \mathbf{W}_i^Q \\ \mathbf{K}_i = \mathbf{x}_E \mathbf{W}_i^K \\ \mathbf{V}_i = \mathbf{x}_E \mathbf{W}_i^V \end{cases}$$

where $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V$ are the parameter matrices of the above-mentioned three mappings.

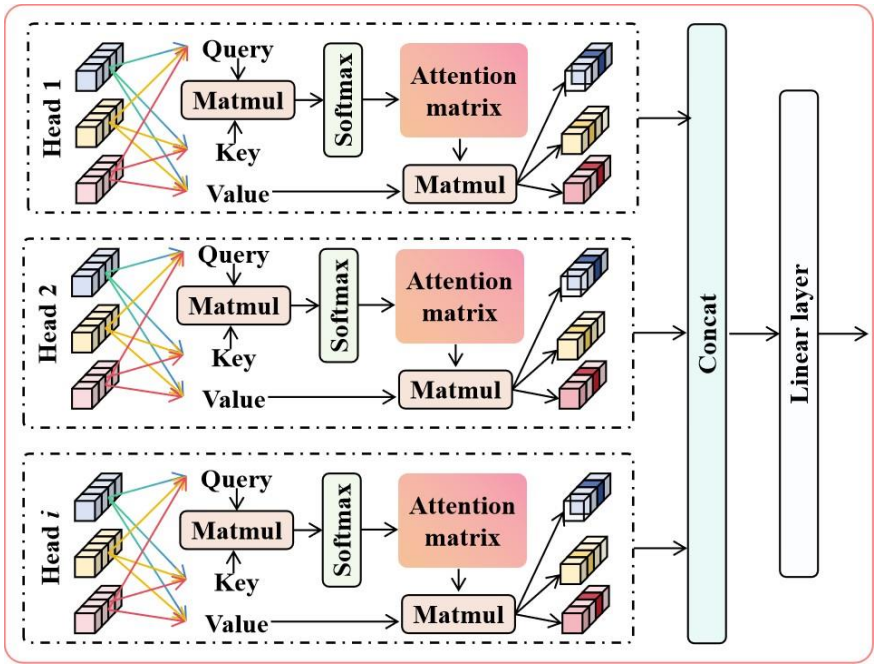


Fig.3 Schematic diagram of the multi-head attention mechanism

The self-attention weight matrix, $\mathbf{A}_i$, is then calculated by querying $\mathbf{Q}_i$ and $\mathbf{K}_i$, which can be expressed as

$$\mathbf{A}_i = \text{softmax}\left(\frac{\mathbf{Q}_i\mathbf{K}_i}{\sqrt{d_K}}\right)$$

where $\text{softmax}(\bullet)$ is the Softmax activation function. And d_K is the sequence dimension of the key vector $\mathbf{K}_i$, expressed as $d_K = \frac{d_m}{m_H}$.

Then, the temporal features captured by the i-th self-attention layer, $\mathbf{H}_i$, can be expressed as

$$\mathbf{H}_i = \mathrm{A}_i\mathbf{V}_i$$

It can be seen that the self-attention weight matrix enhances useful features and weakens useless features. More specifically, the multi-head self-attention mechanism can assign greater weights to important information in the input and smaller weights to useless information. During training, the model focuses more on important information and ignores useless information, thereby improving the recognition of captured features $\mathbf{H}_i$ [41]. Finally, by fusing the m_H features captured by each self-attention layer, the features extracted by the multi-head attention mechanism can be expressed as

$$\mathbf{H}_M = \text{Concat}\left(\mathbf{H}_1, \mathbf{H}_2, \cdots, \mathbf{H}_{m_H}\right)\mathbf{W}^M$$

where $\text{Concat}(\bullet)$ is the tensor concatenation operation; and $\mathbf{W}^M$ is the parameter matrix for linear mapping.

Short-cuts & layer normalizations: The Short-cuts & layer normalizations aim to reduce the

training difficulty of the DSAFAN. As the number of neural network layers continues to increase, the model accuracy on the training set saturates or even decreases (i.e., the degradation problem)[42]. Short-cuts can effectively suppress degradation by introducing multiple cross-layer identity mappings, thereby enhancing the training stability of deep neural networks. Moreover, during the training process of deep neural networks, parameter updates in the preceding layers may change the input distribution in the following layers (i.e., internal covariate shift)[43]. The data distribution can start varying drastically, and hence, the subsequent layers need to constantly adapt to the parameter updates in the preceding layers, reducing the model training speed. Layer standardization can reduce the internal covariate shift between different network layers, thereby accelerating the convergence of deep neural networks.

Fully connected feed-forward layer: The fully connected feed-forward layer improves the nonlinear learning ability of the DSAFAN, which consists of two linear transformation layers with a ReLU activation function. The output of the fully connected feed-forward layer can then be expressed as

$$\mathbf{H}_{FFN} = \max\left\{\mathbf{H}_{AN}\mathbf{W}_1^F + \mathbf{b}_1^F, 0\right\}\bullet\mathbf{W}_2^F + \mathbf{b}_2^F$$

where $\mathbf{H}_{AN}$ is the output of feature representation $\mathbf{H}_M$ after the short-cuts & layer normalizations. $\mathbf{W}_1^F, \mathbf{W}_2^F$ are the parameter matrices of the two linear transformations; and $\mathbf{b}_1^F, \mathbf{b}_2^F$ are the bias vectors of the two linear transformations.

3.4 MOOST-based feature augmentation for inter-class imbalance

The objective of the feature augmentation module is to mitigate inter-class imbalance while improving data completeness. The popular SMOTE synthesizes abnormal samples by randomly selecting points on the line connecting two neighboring abnormal samples. Let us assume that $\mathbf{h}$ is any abnormal sample in the abnormal feature representation set $\mathbf{H}^A$ and $\tilde{\mathbf{h}}$ is a randomly selected abnormal sample from its neighboring fault samples. The newly synthesized fault sample can then be expressed as

$$\mathbf{h}_{New} = \mathbf{h} + \lambda\left(\tilde{\mathbf{h}} - \mathbf{h}\right) = \left(1-\lambda\right)\mathbf{h} + \lambda\tilde{\mathbf{h}}$$

where λ is the interpolation factor ($0 \le \lambda \le 1$) and follows the uniform distribution $\mathrm{U}(0,1)$.

As shown in Fig.4 (a), SMOTE can only synthesize in-distribution abnormal data, which inherently fails to expand the region occupied by existing anomalies, resulting in limited benefits for decision boundary learning. To address this issue, the MOOST modifies the nearest neighbor selection strategy to reduce the risk of synthesized data falling into the danger zone, while generating out-of-distribution synthesis data with abnormal characteristics to enhance the completeness of the abnormal data. Notably, Fig.4 (b) illustrates the synthesis process of the

MOOST, which is described in detail below.

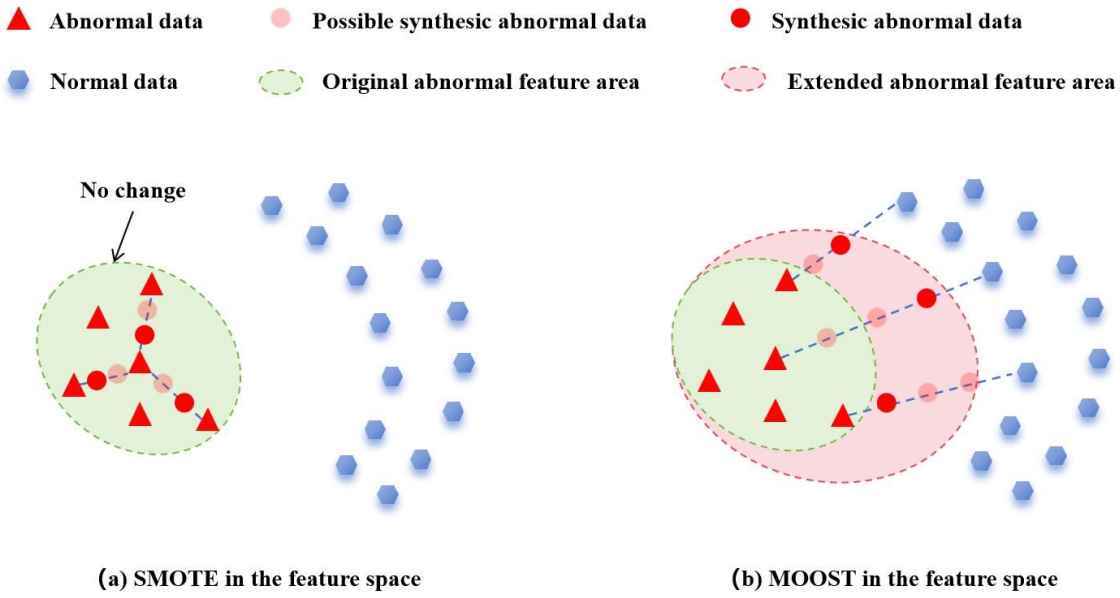


Fig.4 Schematic diagram of the synthesis process of (a) the SMOTE and (b) the MOOST in the feature space

Step 1: The Euclidean distances of the $\mathbf{h}$ with the normal feature representations are calculated, and the normal feature representations are sorted according to their distances, which can be expressed as

$$\mathbf{Z}_{\mathbf{h}}^{N}=\left\{\mathbf{u}_{k}:D_{E}\left(\mathbf{h},\mathbf{u}_{k}\right)\le D_{E}\left(\mathbf{h},\mathbf{u}_{k+1}\right),\mathbf{u}_{k}\in\mathbf{H}^{N}\right\}$$

where $\mathbf{H}^{N}$ is the normal feature representation set, and $D_{E}\left(\mathbf{z},\mathbf{u}_{k}\right)$ is the Euclidean distance between feature representations $\mathbf{h}$ and $\mathbf{u}_{k}$.

Step 2: The first five normal feature representations of $\mathbf{h}$ from the set $\mathbf{Z}_{\mathbf{h}}^{N}$ are selected to form a 5-nearest neighbor set $\mathbf{Z}_{\mathbf{h}5}^{N}$, which can be expressed as

$$\mathbf{Z}_{\mathbf{h}5}^{N}=\left\{\mathbf{u}_{k}:\mathbf{u}_{k}\in\mathbf{Z}_{\mathbf{h}}^{N},k=1,2,3,4,5\right\}$$

Step 3: A feature representation $\mathbf{u}$ is randomly selected from $\mathbf{Z}_{\mathbf{h}5}^{N}$ and combined to form a sample pair $\left[\mathbf{h},\mathbf{u}\right]$. Then, the interpolation factor $\tilde{\lambda}$ is randomly generated using the Beta distribution $Beta\left(\alpha,\beta\right)$. The probability density function of $\tilde{\lambda}$ can be expressed as

$$\varphi\left(\tilde{\lambda};\alpha,\beta\right)=\frac{\Gamma\left(\alpha+\beta\right)}{\Gamma\left(\alpha\right)\Gamma\left(\beta\right)}\tilde{\lambda}^{\alpha-1}\left(1-\tilde{\lambda}\right)^{\beta-1}$$

where $\Gamma\left(\bullet\right)$ is the Gamma function, and α,β represent shape parameters of the Beta distribution.

Step 4: The newly synthesized abnormal feature representation can then be expressed as

$$\mathbf{h}_{New}=\mathbf{h}+\tilde{\lambda}\left(\mathbf{u}-\mathbf{h}\right)=\left(1-\tilde{\lambda}\right)\mathbf{h}+\tilde{\lambda}\mathbf{u}$$

It can be observed that the synthesized data $\mathbf{h}_{New}$ lies outside the region currently occupied by

existing anomalies. Specifically, the MOOST synthesizes minority-class data in an outward direction to expand their advantageous zones, thereby enhancing the generalization capability of deep learning architectures.

Step 5: The process of Step 1 - Step 4 is iteratively applied until an equal number of features across all types is achieved in the training set, thereby generating a balanced feature representation set $\mathbf{H}^B$. It is worth noting that this module is embedded within the deep learning framework, activated during training and deactivated during testing.

3.5 SMC-based feature classification for anomaly detection

After obtaining a balanced feature set in the learning feature space, this paper adopts the SMC as the feature classification module for anomaly classification, which only consists of a linear layer with a Softmax activation function[44]. To learn the mapping relationship between the feature representation and the status label, the binary cross-entropy $\mathcal{L}_{BCE}$ loss is used to optimize the weight parameters of the DSAFAN, which can be expressed by

$$\mathcal{L}_{BCE} = \frac{1}{card\left(\mathbf{H}^B\right)} \sum_{\mathbf{h} \in \mathbf{H}^B} \left[y_{\mathbf{h}} \ln\left(o_{\mathbf{h}}\right) + \left(1 - y_{\mathbf{h}}\right) \ln\left(1 - o_{\mathbf{h}}\right) \right]$$

where $o_{\mathbf{h}}$ is the output of $\mathbf{h}$ though the feature classification module, and $y_{\mathbf{h}}$ is the label of $\mathbf{h}$. Finally, the trained DSAFAN takes the raw data as input and outputs the probability of the data belonging to each category, thereby enabling accurate anomaly classification for aero-engines.

4 Experimental Results

The effectiveness of the proposed anomaly detection method is demonstrated through validation using a real-world operational dataset derived from engines of an Asian airline company.

4.1 Experimental data description

The aero-engine consists of a fan (FAN), a low-pressure compressor (LPC), a high-pressure compressor (HPC), a combustion chamber (CC), a high-pressure turbine (HPT), a low-pressure turbine (LPT), an exhaust system, and various other components. The state characteristic parameters of the aero-engine can reflect the performance status of the entire machine and its unit cells. It is an extremely important step for anomaly detection to extract the abnormal features from the state characteristic parameters. The aero-engine has several state characteristic parameters, among which the most critical ones for anomaly detection include the exhaust gas temperature (EGT), low-pressure rotor speed (N1), high-pressure rotor speed (N2), and fuel flow (FF). However, their measurement values are affected by external environmental factors and operating status, such as temperature and atmospheric pressure. Aiming at this issue, the original equipment

manufacturer (OEM) usually converts measurement values to deviation values according to the external environment and baseline model. To be specific, EGT, N1, N2, and FF are converted into delta exhaust gas temperature (DEGT), delta high-pressure rotor speed (DN2), delta fuel flow (DFF), and exhaust gas temperature margin (EGTM) as the key characteristic parameters (KCPs).

Then, the obtained KCPs are used to construct samples by a sliding window with a step size of 10. As summarized in Table 1, a total of 1017 samples were constructed from the historical operation data of 63 aero-engines, including 954 normal and 63 abnormal samples. Moreover, to reliably demonstrate the effectiveness of the proposed method, five-fold cross validation was applied to partition the obtained dataset; i.e., both the normal and abnormal samples were randomly divided into five subsets, separately, keeping four subsets for training and one for testing each time.

Table 1 Summary of the aero-engine dataset after 5-fold cross-validation

Five-fold cross validation	Training set		Test set	
	Normal	Abnormal	Normal	Abnormal
1st-fold	763	50	191	13
2nd-fold	763	50	191	13
3rd-fold	763	50	191	13
4th-fold	763	51	191	12
5th-fold	764	51	190	12

4.2 Metrics for evaluating anomaly detection results

In the test set, the number of normal samples still far exceeded that of abnormal samples. In this case, relying solely on overall accuracy as an evaluation metric may be misleading. For example, the first-fold test set contained 191 normal samples and only 13 abnormal samples. Even if the anomaly detection model incorrectly diagnoses all test samples as normal, its overall accuracy will remain as high as 93.63%. To better evaluate the diagnostic performance of the anomaly detection models under inter-class imbalance conditions, this paper utilized the confusion matrix (shown in Table 2), true positive rate ($TPR = \frac{N_{11}}{N_{11} + N_{12}}$; the ability of the anomaly detection method to diagnose abnormal data), true negative rate ($TNR = \frac{N_{22}}{N_{21} + N_{22}}$; the ability of the anomaly detection method to diagnose normal data), and balanced accuracy ($BA = \frac{TPR + TNR}{2}$; the overall detection ability of the anomaly detection method) to evaluate the detection capability of the method[45].

Table 2 An illustration of confusion matrix

Actual Class	Predicted Class	
	Abnormal	Normal
Abnormal	N_{11}	N_{12}
Normal	N_{21}	N_{22}

4.3 Hyperparameter settings for experimental methods

All experiments were implemented using Python 3.9 with PyTorch 1.11. The hyperparameters were set according to [17]. First, the feature extraction module included an input & position encoding layer, as well as two Transformer encoder layers, in which $m_H = 4$ and $d_K = 128$. Second, the shape parameters of the Beta distribution in the MOOST were both set to 2. Finally, the feature classification module included a linear output layer with the Softmax activation function. Additionally, the weight parameters of the DSAFAN were optimized using the Adam optimizer with a learning rate of 0.0002 and a batch size of 128.

4.4 Experimental comparative analysis

To verify the detection performance of the proposed method, three comparative experiments were designed. The first experiment served as the Baseline method, which did not utilize any resampling method to balance the training set and directly trained the deep self-attention network (DSANet) on the original imbalanced training set. The DSANet consisted of an input & position encoding layer, two Transformer encoder layers, and a linear output layer with the Softmax activation function. The second group of experiments involved six popular over-sampling techniques for anomaly detection, including SMOTE[14], borderline over-sampling (BOS)[34], safe-level SMOTE (SL-SMOTE)[35], k-means SMOTE (KM-SMOTE)[36], synthetic minority based on probabilistic distribution over-sampling (SymProD)[46], and SMOTE with boosting (SMOTEWB)[47]. The third group of experiments adopted two advanced deep learning-based over-sampling techniques for anomaly detection, including time-series generative adversarial networks (TimeGAN)[48] and augment time-series data augmentation using deep generation models (AugmentTS)[49]. Additionally, to alleviate the randomness of neural networks on the experimental results, each fold experiment was repeated 10 times, and the average value was taken as the final result.

The average TPR, TNR, and BA values of the considered anomaly detection methods were comparatively analyzed. Based on the preceding analysis, TPR serves as a metric for evaluating a method's diagnostic capability in identifying abnormal samples, which was summarized in Table 3. It can be seen that the abnormal sample detection ability of the DSAFAN far exceeded the baseline method and the 8 other over-sampling methods in the five-fold experiments. Specifically,

the DSAFAN demonstrated an average TPR of 89.09%, i.e., a 14.04 % improvement in abnormal sample detection accuracy compared to the SL-SMOTE (second-ranked in terms of the average TPR). In addition, TNR serves as a metric for evaluating a method's diagnostic capability in identifying normal samples, which was shown in Table 4. The detection ability of the DSAFAN underperformed even that of the Baseline in terms of the average TNR. Specifically, the average TNR of DSAFAN was 92.65 %, i.e., a decrease of 6.31% compared to the Baseline method (first-ranked in terms of average TNR). The fact that the TPR of the Baseline was much lower than its TNR indicates that the Baseline was more biased towards normal samples and neglected abnormal samples. Compared to the Baseline, the DSAFAN significantly reduced the gap between the TPR and TNR, effectively alleviating the problems due to inter-class imbalance.

Table 3 Average TPR (%) of different anomaly detection methods

No.	Method	1st-fold	2nd-fold	3rd-fold	4th-fold	5th-fold	Average
1	**Baseline**	86.15	75.38	79.23	59.17	53.33	70.65
2	**SOMTE**	92.31	80.00	73.85	57.50	55.00	71.73
	BOS	92.31	80.77	75.38	54.17	55.83	71.69
	SL-SMOTE	84.62	79.23	73.08	62.50	75.83	75.05
	KM-SMOTE	86.92	79.23	77.69	59.17	57.50	72.10
	SYMPROD	86.92	75.38	73.08	60.00	50.83	69.24
	SMOTEWB	88.46	73.85	79.23	59.17	52.50	70.64
3	**TimeGAN**	86.15	73.85	63.08	55.83	55.83	66.95
	AugmentTS	81.54	70.77	73.85	68.33	63.33	71.56
Developed	**DSAFAN**	92.31	90	92.31	84.17	86.67	**89.09**

Table 4 Average TNR (%) of different anomaly detection methods

No.	Method	1st-fold	2nd-fold	3rd-fold	4th-fold	5th-fold	Average
1	**Baseline**	99.01	98.64	99.58	97.80	99.79	**98.96**
2	**SOMTE**	99.11	98.74	99.32	97.54	99.58	98.85
	BOS	98.85	98.69	99.42	97.80	99.74	98.90
	SL-SMOTE	98.12	97.85	98.95	97.91	98.21	98.20
	KM-SMOTE	98.90	98.53	99.53	97.75	99.47	98.83
	SYMPROD	99.16	98.69	99.58	97.70	98.95	98.82
	SMOTEWB	98.90	98.59	99.37	97.80	99.74	98.88
3	**TimeGAN**	98.90	98.59	99.11	97.02	99.16	98.55
	AugmentTS	99.11	98.69	99.27	97.80	99.89	98.95
Developed	**DSAFAN**	91.88	94.76	92.36	95.03	89.21	92.65

Furthermore, the average BA of the considered anomaly detection methods was summarized in Table 5, which can be concluded as follows. Compared to the Baseline method, 6 out of the 9 considered over-sampling methods exhibited a higher average TPR and BA. And, the average TNR of 8 out of the 9 over-sampling methods was extremely close to that of the baseline. Obviously, appropriate resampling drastically improved the diagnostic performance of anomaly detection methods on imbalanced datasets. Particularly, the average TPR and BA of the SL-SMOTE and KM-SMOTE outperformed those of the SMOTE, which indicated that reducing the probability of generating noisy samples can improve model performance for over-sampling

methods. As depicted in Fig.5, the proposed DSAFAN achieved the best comprehensive detection performance in terms of the average BA. Specifically, compared to the Baseline, SOMTE, BOS, SL-SMOTE, KM-SMOTE, SYMPROD, SMOTEWB, TimeGAN, and AugmentTS, the proposed DSAFAN improved by 6.06%, 5.58%, 5.57%, 4.24%, 5.40%, 6.84%, 6.11%, 8.12%, 5.61%, respectively. While ensuring accurate diagnosis of normal samples, it also exhibited significantly improved detection of abnormal samples. This is mainly because the developed MOOST not only reduced the probability of generating noisy data but also extended the area occupied by the abnormal data, thereby improving the anomaly detection capability of the DSAFAN. Overall, for aero-engines with extremely strict safety requirements, the proposed DSAFAN is the most suitable for anomaly detection.

Table 5 Average BA (%) of different anomaly detection methods

No.	Method	1st-fold	2nd-fold	3rd-fold	4th-fold	5th-fold	Average
1	**Baseline**	92.58	87.01	89.41	78.49	76.56	84.81
2	**SOMTE**	95.71	89.37	86.59	77.52	77.29	85.29
	BOS	95.58	89.73	87.40	75.99	77.79	85.30
	SL-SMOTE	91.37	88.54	86.02	80.21	87.02	86.63
	KM-SMOTE	92.91	88.88	88.61	78.46	78.49	85.47
	SYMPROD	93.04	87.04	86.33	78.85	74.89	84.03
	SMOTEWB	93.68	86.22	89.30	78.49	76.12	84.76
3	**TimeGAN**	92.53	86.22	81.10	76.43	77.50	82.75
	AugmentTS	90.33	84.73	86.56	83.07	81.61	85.26
Developed	**DSAFAN**	92.1	92.38	92.33	89.6	87.94	**90.87**

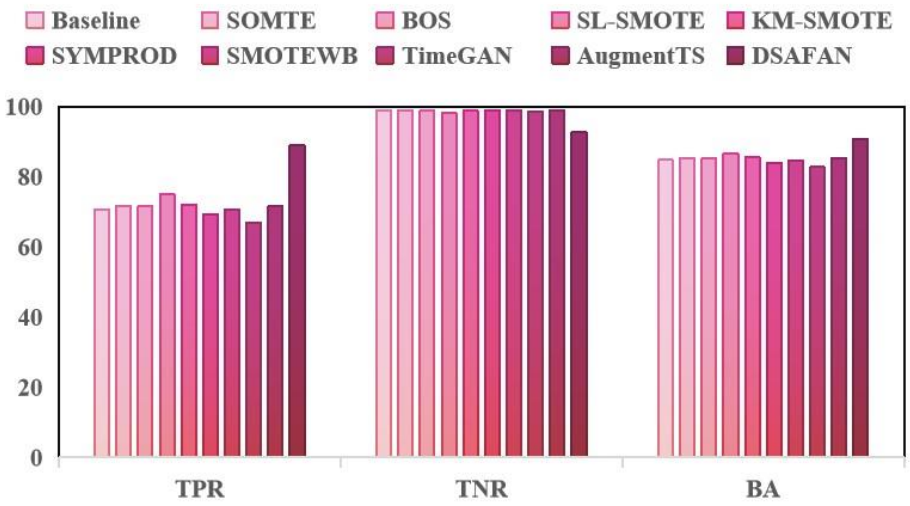


Fig.5 Average TPR (%), TNR (%), and BA (%) of different anomaly detection methods

4.5 Investigation on performance with different distribution expectations

Table 6 - Table 8 showed the average of TPR, TNR, and BA of the proposed DSAFAN using different distribution expectations, respectively. According to the probability density function of the Beta distribution, it can be seen that the distribution expectation represented the extent to which the MOOST extended outward. The greater the distribution expectation, the greater the probability that the synthetic sample is close to the abnormal data, as depicted in Fig. 6. As the

distribution expectation increased, the ability of the DSAFAN to detect abnormal samples gradually strengthened while that of the DSAFAN to detect normal samples gradually weakened, consistent with the above analysis. Moreover, when the distribution expectation is 5/6, the DSAFAN achieved the worst comprehensive detection performance but still outperformed the considered comparison methods, further demonstrating the effectiveness of the proposed method.

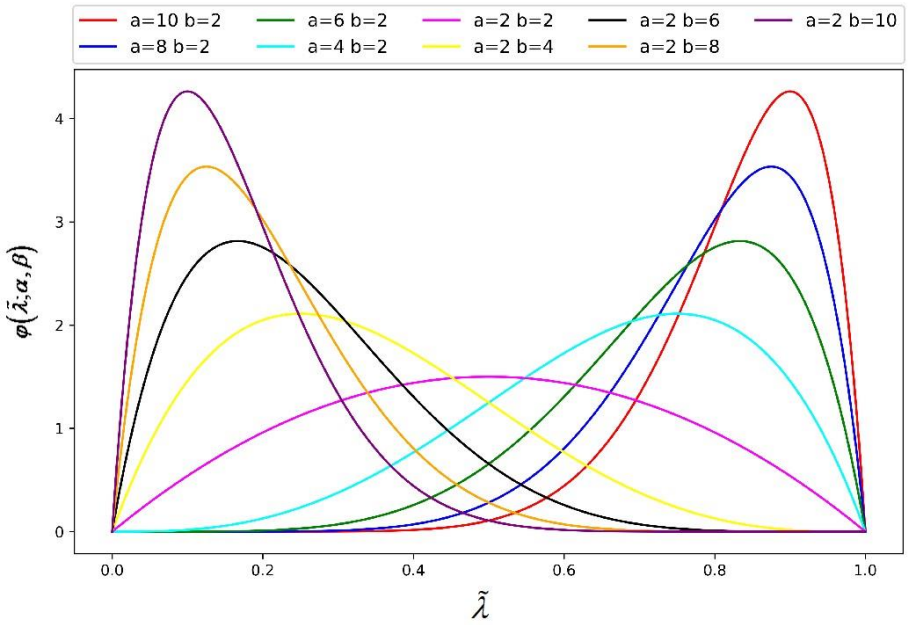


Fig.6 The probability density functions of Beta distributions with different distribution expectations

Table 6 Average TPR (%) of the DSAFAN using different distribution expectations

Expectation	1st-fold	2nd-fold	3rd-fold	4th-fold	5th-fold	Average
$\left(\alpha=10,\beta=2,E=\frac{5}{6}\right)$	93.08	99.23	95.38	91.67	90.83	**94.04**
$\left(\alpha=8,\beta=2,E=\frac{4}{5}\right)$	93.85	98.46	93.85	90	90.83	93.4
$\left(\alpha=6,\beta=2,E=\frac{3}{4}\right)$	93.08	100	93.08	90.83	90.83	93.56
$\left(\alpha=4,\beta=2,E=\frac{2}{3}\right)$	92.31	96.15	89.23	89.17	89.17	91.21
$\left(\alpha=2,\beta=2,E=\frac{1}{2}\right)$	92.31	90	92.31	84.17	86.67	89.09
$\left(\alpha=2,\beta=4,E=\frac{1}{3}\right)$	92.31	87.69	88.46	76.67	83.33	85.69
$\left(\alpha=2,\beta=6,E=\frac{1}{4}\right)$	92.31	85.38	87.69	76.67	76.67	83.74
$\left(\alpha=2,\beta=8,E=\frac{1}{5}\right)$	92.31	84.62	86.15	75.83	74.17	82.62
$\left(\alpha=2,\beta=10,E=\frac{1}{6}\right)$	92.31	84.62	82.31	76.67	75	82.18

Table 7 Average TNR (%) of the DSAFAN using different distribution expectations

Expectation	1st-fold	2nd-fold	3rd-fold	4th-fold	5th-fold	Average
$\left(\alpha=10,\beta=2,E=\frac{5}{6}\right)$	80.58	86.23	81.2	85.03	83.68	83.34
$\left(\alpha=8,\beta=2,E=\frac{4}{5}\right)$	82.57	86.65	85.13	86.49	85.26	85.22
$\left(\alpha=6,\beta=2,E=\frac{3}{4}\right)$	83.77	86.96	86.49	87.75	85.16	86.03
$\left(\alpha=4,\beta=2,E=\frac{2}{3}\right)$	87.02	89.84	88.06	90.79	87.32	88.6
$\left(\alpha=2,\beta=2,E=\frac{1}{2}\right)$	91.88	94.76	92.36	95.03	89.21	92.65
$\left(\alpha=2,\beta=4,E=\frac{1}{3}\right)$	96.28	97.59	94.82	97.28	95.21	96.24
$\left(\alpha=2,\beta=6,E=\frac{1}{4}\right)$	96.96	99.06	96.91	97.23	96.42	97.32
$\left(\alpha=2,\beta=8,E=\frac{1}{5}\right)$	97.17	99.11	97.07	97.75	97.11	97.64
$\left(\alpha=2,\beta=10,E=\frac{1}{6}\right)$	97.12	99.32	97.91	96.49	97.53	**97.67**

Table 8 Average BA (%) of the DSAFAN using different distribution expectations

Expectation	1st-fold	2nd-fold	3rd-fold	4th-fold	5th-fold	Average
$\left(\alpha=10,\beta=2,E=\frac{5}{6}\right)$	86.83	92.73	88.29	88.35	87.26	88.69
$\left(\alpha=8,\beta=2,E=\frac{4}{5}\right)$	88.21	92.56	89.49	88.25	88.05	89.31
$\left(\alpha=6,\beta=2,E=\frac{3}{4}\right)$	88.42	93.48	89.78	89.29	88	89.8
$\left(\alpha=4,\beta=2,E=\frac{2}{3}\right)$	89.66	93	88.65	89.98	88.24	89.9
$\left(\alpha=2,\beta=2,E=\frac{1}{2}\right)$	92.1	92.38	92.33	89.6	87.94	90.87
$\left(\alpha=2,\beta=4,E=\frac{1}{3}\right)$	94.3	92.64	91.64	86.97	89.27	**90.96**
$\left(\alpha=2,\beta=6,E=\frac{1}{4}\right)$	94.64	92.22	92.3	86.95	86.54	90.53
$\left(\alpha=2,\beta=8,E=\frac{1}{5}\right)$	94.74	91.86	91.61	86.79	85.64	90.13
$\left(\alpha=2,\beta=10,E=\frac{1}{6}\right)$	94.71	91.97	90.11	86.58	86.26	89.93

5 Conclusion

To address the inter-class imbalance issue in the cases of inter-class overlap and incomplete abnormal data, this paper proposes a supervised aero-engine anomaly detection method based on the deep self-attention feature augmentation network. Firstly, the proposed DSAFAN employs the multi-head self-attention mechanism to process monitoring data, with the aim of capturing discernible temporal features. Subsequently, under the guidance of normal data, the MOOST in the DSAFAN generates abnormal data within an inter-class separable feature space, thereby not only mitigating the numerical disparity between normal and abnormal data but also augmenting the completeness of the abnormal data. Then, the SMC learns an unbiased decision boundary and achieves accurate aero-engine anomaly detection. Finally, the effectiveness of the DSAFAN is verified through real operation data of aero-engines, and it achieves the best comprehensive detection performance in terms of the average BA. Specifically, compared to the Baseline, SOMTE, BOS, SL-SMOTE, KM-SMOTE, SYMPROD, SMOTEWB, TimeGAN, and AugmentTS, the proposed DSAFAN improved by 6.06%, 5.58%, 5.57%, 4.24%, 5.40%, 6.84%, 6.11%, 8.12%, and 5.61%, respectively. Moreover, imbalanced anomaly detection of high-end equipment is a fast-evolving research field. Future efforts will focus on discovering feature-level over-sampling methods to further improve the accuracy of anomaly detection.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

Data availability

Data will be made available on request.

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